Savings Solution for Improved Financial Management

About the Client

The client is a company offering a savings application designed to help users in **Canada and the United States** achieve their financial goals through **disciplined savings strategies.** They partner with **various well-known businesses** to offer tailored **saving challenges** for employees, providing incentives based on **specific savings goals.**

Client Requirement

The client requested a **savings solution** that would help employees improve their financial habits by offering a seamless way to save money through **challenges and incentives.** The solution needed to support features like **automated savings transfers, rewards,** and **group savings**, allowing users to save together as a group. The goal was to encourage regular savings and provide tools for better financial management.

- A **real-time financial dashboard** displaying key metrics.
- A user-friendly interface for **group savings** and **personalized savings goals.**
- Integration with **banking APIs** for secure financial data aggregation and transfers.
- **Messaging and notification systems** to keep users informed.
- Administrative features for **managing challenges and user participation.**

About the Solution

Product Type : Mobile and web-based savings application.

Overall : The solution integrates **group savings**, personal savings goals, company-sponsored challenges, and automated financial transactions. It also offers tools to **manage group savings,** monitor progress, and track rewards and incentives.

Modules :

1. **Dashboard :** Displays key metrics such as the total amount in the savings vault, user jars, active saving challenges, contests, ballots, and referral rewards. The dashboard also includes financial literacy articles and customer support options.
2. **Saving Challenges & Contest Management :** Users can participate in ongoing saving challenges or create their own "jars" (savings goals), with details on savings amounts, incentives, start and end dates, withdrawal options, and transaction histories. The module also manages Save to Win contests. The new Group Savings feature allows users to collaborate and save together as a group.
3. **Messaging & Notifications :** Centralized communication module providing real-time notifications about saving challenges, user jars, upcoming draws, and personalized messages.
4. **Profile Management :** Users can securely manage personal and banking information, view transaction histories, and adjust preferences for security (e.g., fingerprint authentication).

Platforms :

The **mobile application** is available on both **Android** and **iOS** so that employees can manage their savings.

The **web application**, developed using **Vue.js,** is designed exclusively for admin users.

Microservices in the Platform

The platform was built on a microservices architecture, allowing for modular and scalable development. The microservices included:

Card-Info Service : Dynamically displays dashboard information such as promotional banners, contest details, and user-specific financial data.

Recommender-Service: Recommends saving challenges to users based on their activity and preferences, shown on the dashboard.

Account Service : Manages user accounts and tracks transactions linked to specific savings jars and challenges.

Finance Aggregator Service : Integrates with financial institutions to aggregate bank account data, facilitating secure data exchange and transaction retrieval.

Funds-Transfer Service : Handles all aspects of fund movement, including internal transfers, transaction reconciliation, and bank account linking.

Items-Manager Service : Manages savings jars and challenges, handling operations like creating new challenges, processing contributions, and tracking rewards.

Rule Manager Service : Manages user-defined savings rules, enabling customization of savings strategies.

Rule Execution Service : Automates the execution of savings rules, including fund transfers to savings jars and incentive calculations.

Message-Service : Manages the delivery of dynamic messages, providing users with contextual updates related to their savings challenges and activities.

Notification-Service : Sends real-time notifications, including emails and alerts, to keep users informed about saving progress, challenges, and rewards.

Public-Callback Service : Handles webhook events from third-party providers like Synctera and Plaid, ensuring real-time bank transaction updates are reflected in the app.

Seamless API Integration : Integrated **Plaid API** in Canada and **Synctera API** in the US for secure banking connections, enabling **KYC verification**, ACH payments, and **internal transfers.**

Integrations

Public-Callback Service : Implemented to handle webhook processing from third-party providers like **Synctera, Plaid**, and **Alterna,** ensuring that all external events, such as bank transactions or account updates, are processed in real-time and reflected accurately within the application.

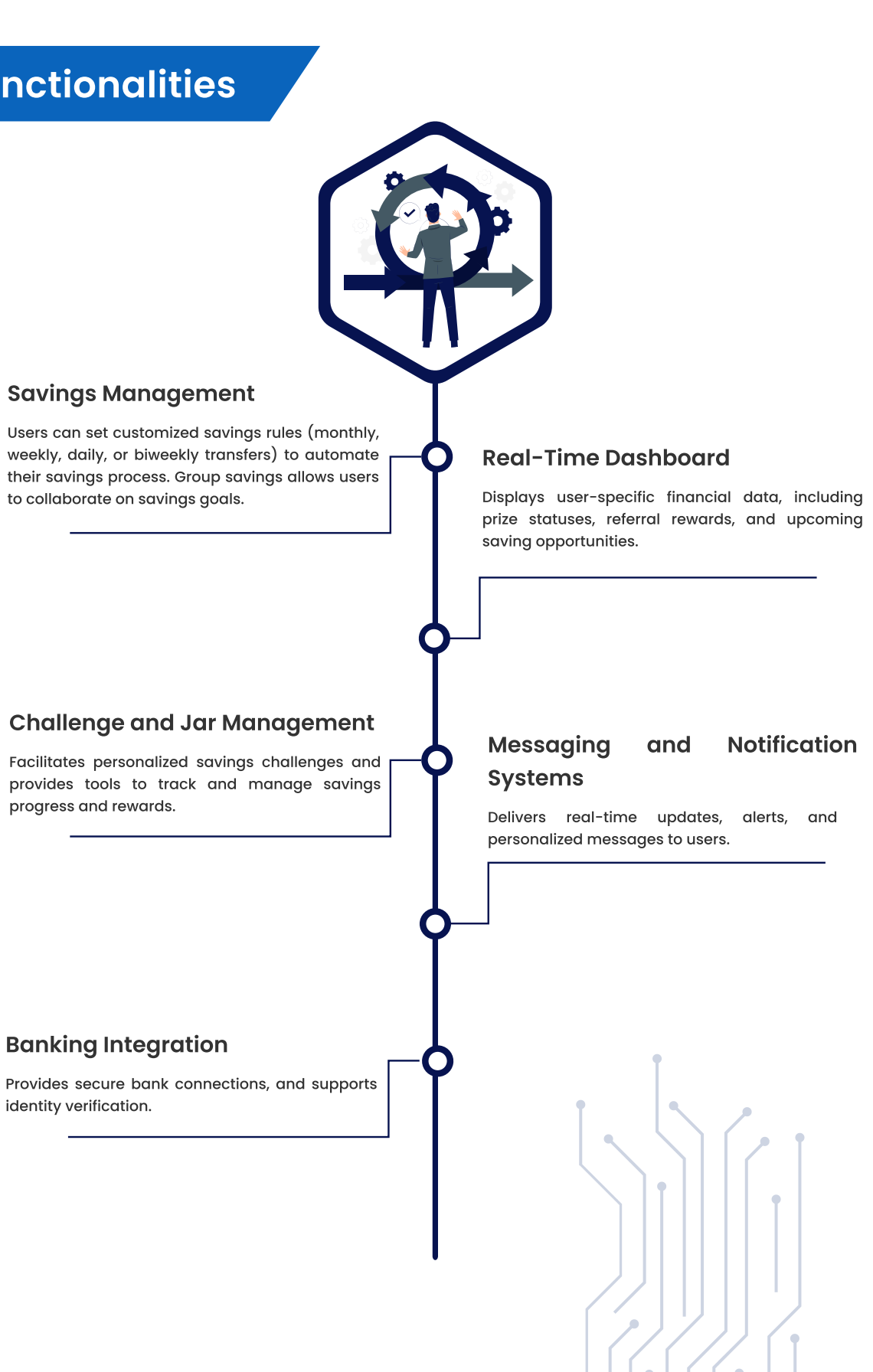
API Integrations

Plaid API (Canada & US) : Facilitates secure and seamless connections between users' bank accounts and the application. Enhances the digital banking experience by providing better risk management, faster account funding, and secure data connectivity.

Synctera API Integration (US) : Supports various banking functionalities essential for the application, including user onboarding with KYC verification, managing internal transfers, processing ACH payments, and handling payment schedules and accounts.

Alterna API (Canada) : Enables users to initiate savings transfers and manage their funds efficiently within the app. It also ensures compliance with Canadian banking standards, offering additional security layers for sensitive financial data.

Functionalities



Savings Management

Users can set customized savings rules (monthly, weekly, daily, or biweekly transfers) to automate their savings process. Group savings allows users to collaborate on savings goals.

Real-Time Dashboard

Displays user-specific financial data, including prize statuses, referral rewards, and upcoming saving opportunities.

Challenge and Jar Management

Facilitates personalized savings challenges and provides tools to track and manage savings progress and rewards.

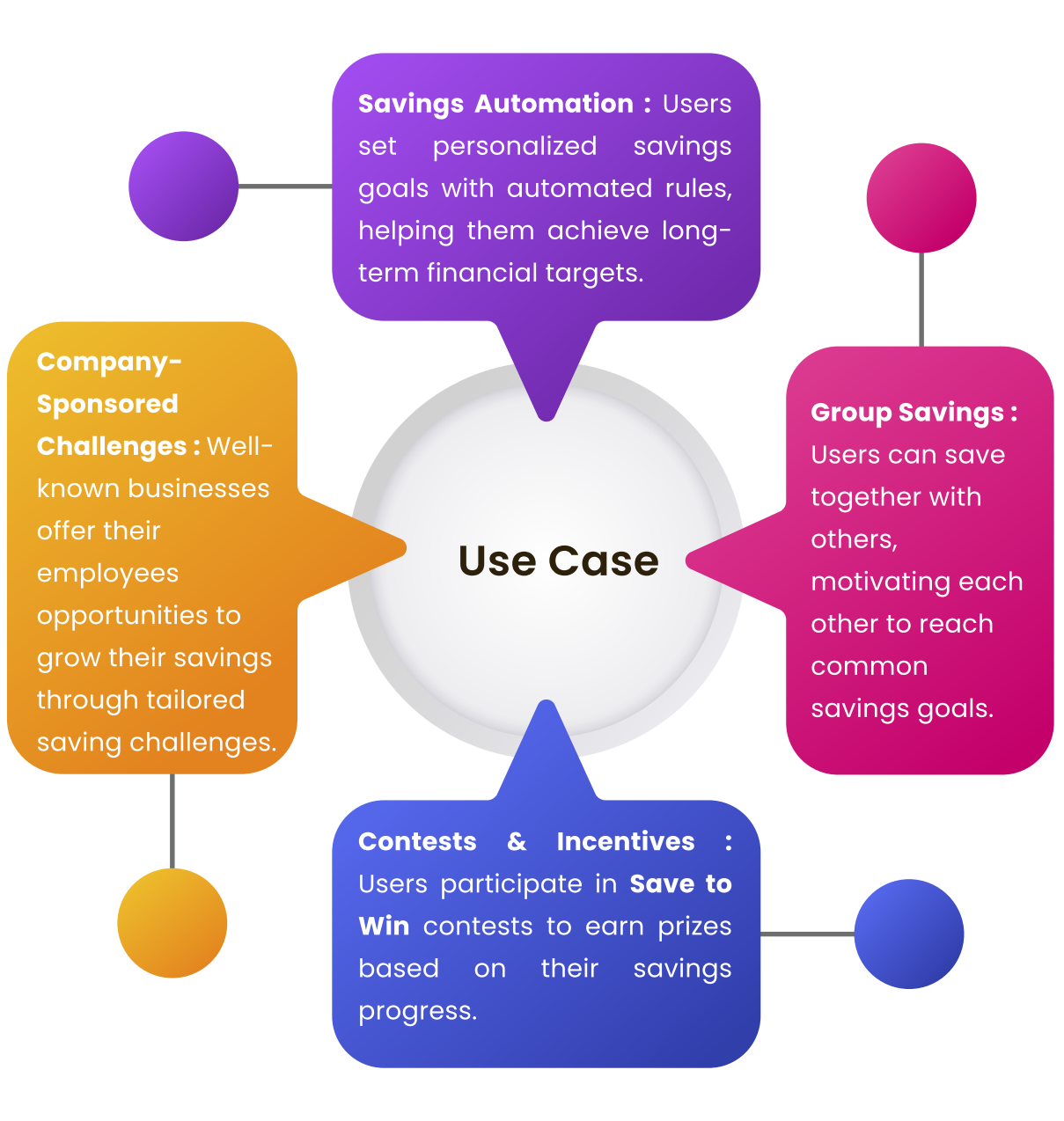
Messaging and Notification Systems

Delivers real-time updates, alerts, and personalized messages to users.

Banking Integration

Provides secure bank connections, and supports identity verification.

Use Cases



Use Case

Savings Automation : Users set personalized savings goals with automated rules, helping them achieve long-term financial targets.

Group Savings : Users can save together with others, motivating each other to reach common savings goals.

Contests & Incentives : Users participate in **Save to Win** contests to earn prizes based on their savings progress.

Company-Sponsored Challenges : Well-known businesses offer their employees opportunities to grow their savings through tailored saving challenges.

Value Proposition

- Solves the problem of **manual savings management** by automating the process through customizable saving rules and goals.
- Encourages user engagement through **savings incentives** offered by employers.
- Provides a platform for **financial education** empowering users to make smarter financial decisions.
- **Improves user efficiency** by automating routine transactions and providing a clear, real-time overview of savings progress.
- Simplifies the process of **secure bank integration,** ensuring user data is safe and accessible.
- **Optimized Performance:** Addressed bottlenecks by enhancing performance, and ensuring identity verification for a better user experience.

Development Challenges and Solutions

During the development of the solution, we encountered several challenges:

DB Lock Problems : System performance was affected by database lock issues. We solved this by **analyzing logs**, identifying the root cause, and implementing changes to prevent future occurrences. Query management was also optimized for better efficiency.

Performance Bottlenecks : Certain processes, such as **scheduler jobs,** took too much time. We optimized the implementation logic, reducing load times and improving overall performance.

End Users

Primary Audience : Consumers in **Canada and the US** looking for tools to improve their financial savings habits.

User Role :

Admin Users : Manage saving challenges, monitor user participation, and oversee contests and rewards.

End-Users : Set savings goals, manage jars, and participate in contests. They can track their progress, monitor financial transactions, and adjust their personal settings.

Advantages :

Improves savings discipline through **automated savings rules and group savings.**

Engages users with **company-sponsored challenges and real-time incentives.**

Provides secure, efficient **bank integration** for easy fund management.

Tech Stack

Front-End:

iOS
Android
Vue.js

Back-End :

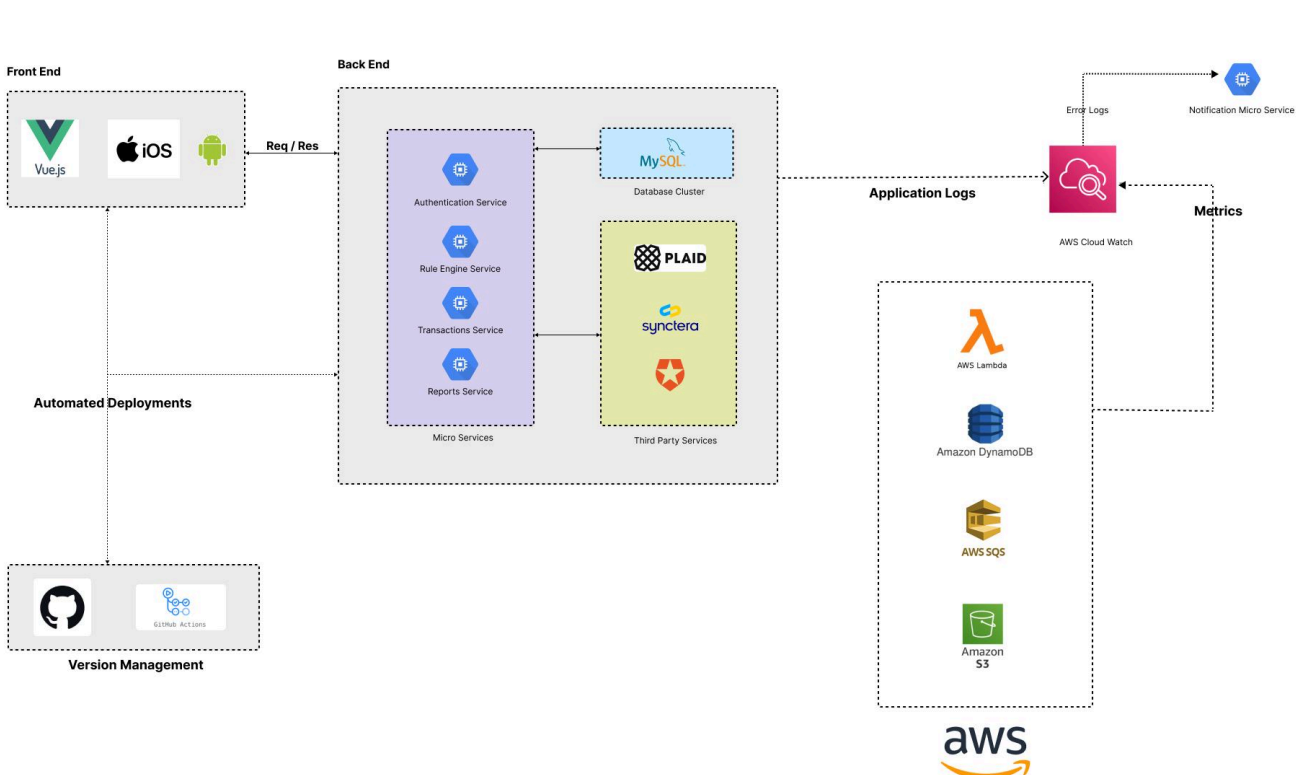
Java
My Sql
AWS EC2
Auth
Plaid API
Synctera

- **Microservices Architecture :** Ensured scalability and modularity for the platform, allowing independent updates to different parts of the system.
- **Data-Common Service :** Developed to handle shared entities and repositories across microservices, ensuring smooth integration and consistency across services.
- **Parent Service :** Leveraged to handle common functionalities like **authentication (Auth0), file management (AWS S3), messaging (Mailgun),** and a common class for **Synctera**, which contains the details related to the API, model, and authentication.

Support & Maintenance

- **Ongoing Maintenance :** Regular monitoring of system performance and user feedback.
- **Updates :** Continuous rollout of new features and improvements based on client and user feedback.
- **Security :** Ensured the platform remained compliant with financial regulations and security standards like **KYC**.
- **Infrastructure Monitoring :** Deployed on **AWS Cloud** with performance tracking using **CloudWatch** to ensure optimal reliability and uptime.
- **Post-Deployment Support :** Provided technical support for bug fixes, updates, and the addition of new features based on evolving business needs.

Architecture Design



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